SURA Mid Term Evaluation Report

# Machine Learning-Based User Localization Using Hybrid Magnetic Sensor and Wi-Fi Received Signal Strength (RSS) Data in GPS-Denied Scenarios

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**Mentor**
Prof. Neel Kanth Kundu
Centre for Applied Research in Electronics (CARE),
Indian Institute of Technology Delhi

**Students**
Utkarsh Agrawal (2024CS10076)
Jayendra Vijay Birhade (2024CS10891)
Department of Computer Science and Engineering,
Indian Institute of Technology Delhi



**Abstract**

This report documents mid-term progress on the SURA Indoor Positioning project, which aims to learn a static spatial fingerprint of an indoor environment for real-time, on-device localization [1, 2]. We describe the core system constraints — causal inference, cross-device generalization, and fingerprint-based rather than trajectory-based learning — and trace the evolution of five model architectures from an initial mean absolute error (MAE) of 6.36 m down to a current best of 1.43 m. We formally identify four fundamental flaws in the initial experimental setup and present the redesigned three-part causal system, comprising a WiFi heatmap environment model [3, 4], causal pedestrian dead-reckoning [5, 6], and a KalmanNet-based filter [7], that resolves each flaw. The reported 1.43 m MAE represents a 2.5× improvement over the strongest prior model.

# 1. Problem Statement and System Constraints

The central challenge of this project is to learn an environment fingerprint — a static, spatially unique, and temporally stable characterization of a physical space — from heterogeneous sensor streams (WiFi RSSI and IMU) [1, 8]. The critical distinction from trajectory-based approaches [9, 10] is that the system must learn location identity, not path-specific dynamics. This means the same model must localize a user correctly regardless of which direction they are walking or which session they are in.

Five hard constraints govern every design decision in this project.

| Constraint | Requirement | Architectural implication |
|---|---|---|
| Causal inference | No look-ahead; strictly real-time | Bidirectional RNNs are excluded |
| On-device deployment | Full pipeline runs locally on the handset | No server round-trips at inference time |
| Cross-device generalization | Works on unseen phones without retraining | Per-device calibration is not a viable approach |
| Cross-session generalization | Works in walk sessions not seen during training | Route memorization must be eliminated by design |
| Ground truth fidelity | Per-node labels from the static database are preferred | Fabricated or interpolated labels are rejected |

Table 1. Hard constraints governing all architectural choices.

A secondary structural difficulty is spatial multi-modality. A single WiFi or magnetic fingerprint measurement can correspond to multiple physical positions, particularly along corridor segments where sensor signatures look very similar [3]. A model that emits a single (X, Y) coordinate cannot represent this uncertainty and will systematically pull its predictions toward the centroid of the candidate set. This observation motivates the shift to probabilistic heatmap outputs described in Section 4.

# 2. Initial Model Architectures

Four model architectures were developed and benchmarked during the initial phase of the project. All four were trained on continuous-walk CSV files (Continuous_Fused_*.csv). As documented in Section 3, these files contain fundamental data validity issues; the figures here are reported as originally measured, with the full caveats addressed below.

## 2.1 CNN-LSTM (Non-Causal)

Implemented in train_cnn_lstm.py. A one-dimensional convolutional encoder was paired with a bidirectional LSTM [9] to capture both local sensor patterns and long-range temporal context. The reported MAE was 5.04 m. The bidirectional LSTM is, however, a fundamental causality violation: it accessed future sensor readings to infer the current position, making the model deployment-invalid regardless of its accuracy.

## 2.2 Causal Hybrid

Implemented in train_causal_hybrid.py. The bidirectional LSTM was replaced with a unidirectional variant and all convolutions were made causal, achieving full deployment compliance. This model produced the strongest initial result at 3.64 m MAE. Post-hoc analysis revealed that the accuracy arose from route memorization rather than genuine spatial fingerprinting.

## 2.3 Environment Model

Implemented in train_env_model.py. This model was designed explicitly to learn per-location signatures rather than temporal sequences [4]. It achieved 6.36 m MAE on the forward corridor path, but collapsed to 63 m MAE when the same path was walked in reverse. The root cause is that raw magnetometer axes (Mag_x, Mag_y, Mag_z) are measured in the phone body frame and rotate with heading [11], so the same physical spot yields different axis values depending on direction of travel. The model had learned a heading-conditioned shortcut, not a true spatial fingerprint.

### 2.4 Neural EKF with Learned Alpha-Gate

Implemented in train_ekf_fusion.py. This model introduced a three-stage learned fusion design. A spatial anchor branch combined WiFi and magnetic MLPs to produce a per-frame estimate P_spatial. A motion tracker built from causal convolutions and a unidirectional LSTM predicted step deltas. A sigmoid-gated alpha value then blended the two signals. The composite training loss applied weights of 1.0 on the final output, 0.2 on the spatial branch, and 2.0 on the motion branch. At inference the update rule was: P_final[t] = alpha[t] × P_spatial[t] + (1 − alpha[t]) × (P_final[t−1] + delta[t]). Reported MAE: 4.29 m. A mean alpha near 0.6 indicates a slight preference for the spatial anchor. Route memorization was not eliminated.

| Model | MAE | Causal | Route memory risk | Primary issue |
|---|---|---|---|---|
| CNN-LSTM | 5.04 m | No | High | Bidirectional LSTM; deployment-invalid |
| Causal Hybrid | 3.64 m | Yes | High | Memorizes the route, not the space |
| Environment Model | 6.36 m / 63 m * | Yes | Medium | Direction-dependent magnetometer features |
| Neural EKF | 4.29 m | Yes | Medium | Learned fusion; still route-biased |

Table 2. Summary of initial architectures. * Forward path / reversed path.

## 3. Identification of Fundamental Experimental Flaws

A systematic audit of the initial experimental pipeline identified four independent sources of invalidity. Each flaw on its own is sufficient to make the reported figures unreliable; taken together they mean that the initial results measured something other than true indoor localization performance.

### 3.1 Single one-dimensional training trajectory

All four Continuous_Fused_*.csv files trace the same corridor: the same start point at approximately (90, 24), the same end region, and only two turns along the entire path. The dataset yielded roughly 325 training windows and 99 test windows, all drawn from a single narrow line through a two-dimensional building floor. No model trained on this data could learn a generalizable fingerprint; it could only memorize the one trajectory it was shown.

### 3.2 Direction-dependent magnetometer axes

Raw Mag_x, Mag_y, and Mag_z are measured in the phone body frame and rotate with heading [11]. The same physical location therefore produces different axis values depending on which direction the user is walking. Model 3's collapse from 6.36 m to 63 m MAE on path reversal is a direct consequence — the model had learned a heading-conditioned shortcut that failed the moment heading changed.

### 3.3 Single-point regression and multi-modal ambiguity

Several locations along a corridor appear similar in sensor space [3, 8]. A regression model forced to emit one (X, Y) coordinate cannot represent the resulting ambiguity — instead of outputting a distribution over candidate locations, it collapses to a centroid. This is a fundamental limitation of the target formulation, not a hyperparameter problem, and it systematically degrades accuracy wherever the environment is spatially ambiguous.

### 3.4 Fabricated ground truth

The Continuous_Fused_*.csv files were generated by fuse_continuous_wifi.py, which used BE Building coordinates rather than the IT Engineering layout, linearly interpolated True_X and True_Y by timestamp from the ordered static-node list, and simulated WiFi measurements by injecting Gaussian noise around static node scans. The ground truth was not independently observed — it was constructed from the same database used to generate the model inputs. Models that learned the node sequence appeared accurate precisely because the labels encoded that sequence. The reported errors reflected consistency with a fabricated trajectory, not validity against real-world positions.

## 4. Redesigned Architecture

The redesigned system resolves all four documented flaws by separating the problem into components that can be trained and validated independently on the static fingerprint database [1, 8], rather than on continuous walks. The static database provides 538 node visits across 167 of 168 nodes, giving genuine two-dimensional spatial coverage with independently observed ground truth.

### 4.1 Design rationale

Three decisions follow directly from the flaw analysis. The environment model is trained only on static node visits, which eliminates flaws 1 and 4. It outputs a probability distribution over a one-metre grid rather than a single coordinate, which eliminates flaw 3. All features are computed from heading-invariant quantities — specifically, the magnitude |M| and per-frame WiFi RSSI vectors [3, 11] — which eliminates flaw 2. The fusion stage uses KalmanNet [7] to keep nearly zero route memory while learning context-dependent gain matrices.

### 4.2 System components

| Component | Role | Key property |
|---|---|---|
| WiFi heatmap environment model | Outputs a probability distribution over a 1 m grid per frame. The spatial spread of the heatmap becomes the Kalman measurement covariance R. | Trained only on static DB. Per-frame and direction-invariant. [3, 4] |
| Online magnetometer normalizer | Fits an ellipsoid from a trailing buffer of raw IMU samples. Outputs calibrated magnitude \|M\| rather than raw body-frame axes. | 3× cross-device spread reduction: 1.77 to 0.57 μT. [11] |
| Causal PDR (Pedestrian Dead-Reckoning) | Step detection from accelerometer magnitude. Predicts a motion delta for each detected step. | Strictly causal. Approx. 1.5 m MAE on short walks. [5, 6] |
| KalmanNet GRU Fusion | Recurrent neural filter replacing fixed Kalman gain with a learned 2×2 matrix gain computed by a GRU. No linear-Gaussian assumption. | No route memory. Fully causal. Adaptive gain from context. [7] |

Table 3. Components of the redesigned four-part causal system.

### 4.3 Inference pipeline

At inference time the pipeline is fully sequential and causal. Raw IMU samples enter the online magnetometer normalizer, which maintains a trailing buffer and fits an ellipsoid to produce a calibrated |M| value consistent across devices [11]. Per-frame sensor readings then enter the WiFi heatmap environment model, which outputs a probability distribution over the spatial grid [3, 4]. KalmanNet [7] fuses this distribution — using its spatial spread as the adaptive measurement covariance and computing a per-step learned gain — with the PDR motion prediction to produce a refined position estimate [5, 6]. No future observations are required at any step and no memory of the walked path is retained.

# 5. Results

All results are reported under two evaluation protocols. The random split assesses generalization to unseen visits of nodes seen during training. The held-out device split uses a Samsung Galaxy S9+ that was excluded entirely from training, testing cross-device generalization.

## 5.1 WiFi heatmap environment model

| **Protocol** | **MAE** | **vs. best prior (3.64 m)** | **What it measures** |
|---|---|---|---|
| Random split (unseen visits) | 1.43 m | 2.5× improvement | Generalizes to unseen visits of known nodes |
| Held-out device (Samsung Galaxy S9+) | 2.02 m | 1.8× improvement | Generalizes to a completely unseen handset |

Table 4. WiFi heatmap environment model performance.

## 5.2 Online magnetometer normalizer

The normalizer was validated by running calibration on all four continuous walk recordings and measuring the standard deviation of |M| across devices before and after [11]. Cross-device spread dropped from 1.77 μT to 0.57 μT, a reduction of approximately 3×, enabling a single environment model to work across different handsets without any per-device calibration step.

## 5.3 Causal PDR

The causal PDR [5, 6] achieves approximately 1.5 m MAE on short walks. Drift accumulates on longer paths, as expected for any dead-reckoning system. In the full KalmanNet pipeline the WiFi heatmap anchor corrects this drift roughly every one second, preventing unbounded error growth.

## 5.4 MAE progression across all models

| **Model** | **MAE** | **Training data** | **Output** | **Route memory** |
|---|---|---|---|---|
| CNN-LSTM | 5.04 m | Continuous walks | Single (X, Y) | High |
| Causal Hybrid | 3.64 m | Continuous walks | Single (X, Y) | High |
| Environment Model | 6.36 m / 63 m * | Continuous walks | Single (X, Y) | Medium |
| Neural EKF | 4.29 m | Continuous walks | Single (X, Y) | Medium |
| **WiFi Heatmap + KalmanNet (current)** | 1.43 m / 2.02 m + | Static fingerprint DB | Probability heatmap | None |

Table 5. Full MAE evolution across all five models. * Forward / reversed path. + Random split / held-out device. The highlighted row is the current system.

# 6. Path Forward

The WiFi heatmap environment model is validated and ready. The KalmanNet fusion pipeline [7] is implemented and individually verified on physically faithful synthetic trajectories [12]. The single remaining prerequisite for end-to-end system validation is the collection of continuous walk recordings with per-frame independent ground truth across the IT Engineering building — explicitly not synthesized by the method described in Section 3.4.

| Step | Action | Dependency |
|---|---|---|
| 1 | Collect WiFi-enabled walks with independent per-frame ground truth | Physical access and logging setup |
| 2 | Validate the complete KalmanNet pipeline end-to-end on the collected data | Step 1 |
| 3 | Evaluate on held-out devices and unseen sessions | Step 2 |
| 4 | Deploy on-device and benchmark inference latency | Step 3 |

Table 6. Planned steps toward full system deployment.

Once Step 1 is complete the system is architecturally finished and will require only validation. No further redesign is anticipated unless real-world data reveals environmental characteristics not captured in the static fingerprint database.

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